

Sid Mathur's Thematic ETF Screener – Daily Report

2026-06-13

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Action Center

Run date: 2026-06-13 **Universe:** 660 ETFs / 26 clusters

Full PC1: FLAT **Clusters long:** 5/26 **ETFs long:** 211/660 (32%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

No new cluster crossovers today.

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_14	Financials	2.491	1	1.00	1.000	1.00
cluster_8	U.S. Dividend Equity	2.474	1	1.00	1.000	1.00
cluster_2	U.S. Mid Cap Equity	3.554	1	1.00	1.000	0.98
cluster_15	Real Estate	2.723	1	0.99	1.000	0.96
cluster_3	U.S. Large Cap Value	2.819	1	1.00	0.991	0.86

Imminent SELL (top 3 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_9	Asia-Pacific Equity	-1.207	1.000	0.99	-1.000	-0.99
cluster_25	International Fixed Income	-0.439	1.000	0.99	-1.000	-0.99
cluster_5	Aggregate Fixed Income	0.035	0.343	0.36	-0.332	-0.24

Full Sample PC1

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 140.671 20dma: 141.019 50dma: 138.859

Spread vs 50dma: +1.813 Spread vs 20dma: -0.348

P(buy crossover within 5 days): 74.7% P(sell): 83.7%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VHT	cluster_11	Healthcare	1.000	26	3.2%	12.77%	0.93
MUB	cluster_16	Municipal Bond	0.881	11	0.1%	18.09%	0.95
BND	cluster_5	Aggregate Fixed Income	-0.243	10	-0.3%	20.14%	0.99
BBAX	cluster_9	Asia-Pacific Equity	-0.986	14	0.1%	6.11%	0.97
IGF	cluster_9	Asia-Pacific Equity	-0.986	14	0.1%	3.90%	0.97
EWZ	cluster_9	Asia-Pacific Equity	-0.986	14	0.1%	1.27%	0.97
BWX	cluster_25	International Fixed Income	-0.988	10	-1.0%	15.01%	0.93

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
VTV	cluster_3	U.S. Large Cap Value	1.00	0.863	1.098	0.97
XLF	cluster_14	Financials	1.00	1.000	1.097	0.96
IJH	cluster_2	U.S. Mid Cap Equity	1.00	0.980	1.018	0.99
SCHD	cluster_8	U.S. Dividend Equity	1.00	0.999	0.894	0.96
VNQ	cluster_15	Real Estate	0.99	0.959	0.971	0.99
QQQ	cluster_6	U.S. Large Cap Growth	0.92	-0.050	1.036	0.98

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
11	cluster_11	Healthcare	Healthcare	17	LONG	shifted	26	3.2%	1.000	+2.000
14	cluster_14	Financials	Financials	14	FLAT	shifted			1.000	+2.000
8	cluster_8	U.S. Dividend Equity	U.S. Dividend Equ...	25	FLAT	shifted			0.999	+0.007
2	cluster_2	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	71	FLAT	shifted			0.980	+1.801
15	cluster_15	Real Estate	Real Estate	14	FLAT				0.959	+0.059
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		11	0.1%	0.881	+0.000
3	cluster_3	U.S. Large Cap Value	U.S. Large Cap Va...	61	FLAT	shifted			0.863	+1.380
18	cluster_18	Consumer Staples	Consumer Staples	11	FLAT				0.108	+0.422
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			0.013	-0.005
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	56	FLAT	shifted			-0.050	+0.188
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	LONG	shifted	10	-0.3%	-0.243	-1.196
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				-0.244	+0.127
1	cluster_1	U.S. Large Cap Equity	U.S. Large Cap Eq...	80	FLAT	shifted			-0.316	-1.164
7	cluster_7	Emerging Markets Equity	Emerging Markets ...	25	FLAT	shifted			-0.490	-0.471
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	61	FLAT	shifted			-0.496	-0.073
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.527	+0.191
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT	shifted			-0.803	+0.197
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.855	-0.034
9	cluster_9	Asia-Pacific Equity	Asia-Pacific Equi...	23	LONG	shifted	14	0.1%	-0.986	-0.282
25	cluster_25	International Fixed Income	International Fix...	6	LONG		10	-1.0%	-0.988	+0.011
19	cluster_19	India Equity	India Equity	11	FLAT				-0.996	+0.004
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.999	+0.001
12	cluster_12	Materials	Materials	17	FLAT	shifted			-1.000	-0.069
13	cluster_13	China Equity	China Equity	15	FLAT	shifted			-1.000	-2.000
17	cluster_17	Gold	Gold	12	FLAT				-1.000	+0.000
20	cluster_20	Broad Commodities	Broad Commodities	10	FLAT				-1.000	-0.000

Cluster Details

Cluster 1 – U.S. Large Cap Equity

Members: 80 **Var explained:** 0.860 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +28.6%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +23.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1205
U.S. Total Market Equity	7	0.0941
Global Equity	7	0.0923
Consumer Discretionary	6	0.0705
Technology	5	0.0610
Asset Allocation	4	0.0489
Industrials	4	0.0419
U.S. Equity	3	0.0393

U.S. Multifactor Equity	3	0.0392
U.S. Mega Cap Equity	3	0.0392
U.S. Quality / Factor Equity	3	0.0389
U.S. Mid Cap Growth	3	0.0386
U.S. Momentum Equity	3	0.0352
Communication Services	3	0.0347
U.S. ESG Equity	2	0.0267
Momentum Equity	2	0.0243
Commodity	1	0.0134
Large Cap Value Equities	1	0.0132
Large Cap Blend Equities	1	0.0132
Dividend Equity	1	0.0130
U.S. Dividend Equity	1	0.0127
Growth Equity	1	0.0126
IPO Equity	1	0.0120
U.S. Mid Cap Momentum Equity	1	0.0119
Alternatives	1	0.0117
U.S. Small Cap Momentum Equity	1	0.0114
International IPO Equity	1	0.0105
Israel Equity	1	0.0099
Singapore Equity	1	0.0095

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
VOO	1.01	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
49	IWV	0.0135	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
68	ITOT	0.0135	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
70	SCHB	0.0135	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
74	VTI	0.0135	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
6	SPY	0.0134	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
14	URTH	0.0134	iShares MSCI World ETF	Global Equity	FLAT
45	JHML	0.0134	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
50	IYY	0.0134	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
55	ESGU	0.0134	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
56	IWB	0.0134	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
61	VTHR	0.0134	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT

62	GSLC	0.0134	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
64	VONE	0.0134	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
66	VV	0.0134	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
67	ILCB	0.0134	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
69	IVV	0.0134	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
71	SCHX	0.0134	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
72	SPTM	0.0134	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
73	VOO	0.0134	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
9	ACWI	0.0133	iShares MSCI ACWI ETF	Global Equity	FLAT
15	CRBN	0.0133	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
17	VT	0.0133	Vanguard Total World Stock Index ETF	Global Equity	FLAT
36	TILT	0.0133	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
48	SUSA	0.0133	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
51	LRGF	0.0133	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
53	SPYX	0.0133	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
57	IWL	0.0133	iShares Russell Top 200 ETF	U.S. Equity	FLAT
65	MGC	0.0133	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
10	AOA	0.0132	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
16	SPGM	0.0132	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
47	DSI	0.0132	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
63	EPS	0.0132	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
52	OEF	0.0131	iShares S&P 100 Fund	U.S. Mega Cap Equity	FLAT
58	QUAL	0.0131	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
35	IMCG	0.0130	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
43	QDF	0.0130	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
34	VOT	0.0129	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
44	CATH	0.0129	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT
59	QUS	0.0129	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
13	AOR	0.0128	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
28	IWP	0.0128	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
42	IOO	0.0128	iShares Global 100 ETF	Global Equity	FLAT
54	XLG	0.0128	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
60	SPHQ	0.0128	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
20	XT	0.0127	iShares Future Exponential Technologies...	Technology	FLAT
22	FTC	0.0127	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
41	SPHB	0.0127	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
46	DGRW	0.0127	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
26	RPG	0.0126	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
80	OMFL	0.0126	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
25	IYC	0.0125	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
4	ROBO	0.0124	ROBO Global Robotics and Automation Ind...	Technology	FLAT
23	PDP	0.0124	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
30	VCR	0.0123	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
33	FDIS	0.0123	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
76	VFMO	0.0122	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG

1	XLY	0.0121	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
3	BOTZ	0.0121	Global X Robotics & Artificial Intellig...	Technology	FLAT
12	AOM	0.0120	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
24	FPX	0.0120	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
29	MTUM	0.0120	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
18	FV	0.0119	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
27	XMMO	0.0119	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
7	QAI	0.0117	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
31	VOX	0.0117	Vanguard Communication Services ETF	Communication Services	FLAT
32	FCOM	0.0117	Fidelity MSCI Communication Services In...	Communication Services	FLAT
37	DWAS	0.0114	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
2	XLC	0.0113	State Street Communication Services Sel...	Communication Services	FLAT
79	PRN	0.0113	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
19	PRNT	0.0112	3D Printing (The) ETF	Technology	FLAT
78	RTH	0.0112	VanEck Retail ETF	Consumer Discretionary	FLAT
11	AOK	0.0110	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT
75	QMOM	0.0110	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
21	FPXI	0.0105	First Trust International Equity Opport...	International IPO Equity	FLAT
38	PPA	0.0104	Invesco Aerospace & Defense ETF	Industrials	LONG
40	XAR	0.0104	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
5	ESPO	0.0103	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
77	EIS	0.0099	iShares MSCI Israel ETF	Israel Equity	FLAT
39	ITA	0.0097	iShares U.S. Aerospace & Defense ETF	Industrials	LONG
8	EWS	0.0095	iShares MSCI Singapore ETF	Singapore Equity	FLAT

Cluster 2 – U.S. Mid Cap Equity

Members: 71 **Var explained:** 0.901 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +39.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +27.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1459
U.S. Mid Cap Equity	6	0.0874
U.S. Small Cap Value	6	0.0871
U.S. Small Cap Growth	6	0.0852
Consumer Discretionary	4	0.0527
Industrials	4	0.0505
U.S. Mid Cap Value	3	0.0438
U.S. Mid Cap Growth	3	0.0426
Leveraged / Inverse	2	0.0280
Dividend Equity	2	0.0277
U.S. Small Cap Low Volatility	2	0.0266
U.S. Small Cap Fundamental Equity	1	0.0148
U.S. Small Cap ESG Equity	1	0.0147
Unknown	1	0.0147
U.S. Small/Mid Cap Fundamental Equity	1	0.0147
U.S. Mid/Small Cap Equity	1	0.0146
Mid Cap Value Equities	1	0.0146
U.S. Small Cap Multifactor Equity	1	0.0146
Small Cap Value Equities	1	0.0144
U.S. Small/Mid Cap Equity	1	0.0143
U.S. Mid Cap Dividend Equity	1	0.0143
U.S. Extended Market Equity	1	0.0142

Value Equity	1	0.0142
U.S. Small Cap Dividend Equity	1	0.0141
U.S. Equity	1	0.0141
U.S. Mid Cap Quality Equity	1	0.0141
Growth Equity	1	0.0140
U.S. Small Cap Momentum Equity	1	0.0138
U.S. Small Cap Industrials	1	0.0138
U.S. Micro Cap Equity	1	0.0136
U.S. Small Cap Consumer Discretionary	1	0.0133
U.S. Dividend Equity	1	0.0133
U.S. Mid Cap Low Volatility	1	0.0127
Real Estate	1	0.0114

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
IJH	1.018	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
47	FNDA	0.0148	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
11	NUSC	0.0147	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
12	PRFZ	0.0147	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
15	MDY	0.0147	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
23	IVOO	0.0147	Vanguard S&P Mid-Cap 400 ETF		FLAT
24	VI00	0.0147	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
27	IJR	0.0147	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
30	IJH	0.0147	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
31	SPMD	0.0147	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
32	SPSM	0.0147	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
33	VB	0.0147	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
34	SCHA	0.0147	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
59	VBR	0.0147	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
8	FNX	0.0146	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
9	FYX	0.0146	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
13	SMLF	0.0146	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
16	IJT	0.0146	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
21	SLYG	0.0146	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
44	EZM	0.0146	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
50	IWN	0.0146	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT

52	IJJ	0.0146	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
54	IVOV	0.0146	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
55	MDYV	0.0146	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
4	IWM	0.0145	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
5	JHMM	0.0145	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
7	SCHM	0.0145	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
18	VIOG	0.0145	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
25	VTWO	0.0145	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
58	VTWV	0.0145	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
64	RWK	0.0145	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
43	EES	0.0144	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
45	PSC	0.0144	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
53	IJS	0.0144	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
56	SLYV	0.0144	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
57	VIOV	0.0144	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
60	RWJ	0.0144	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
70	SDVY	0.0144	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
42	DON	0.0143	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
71	ISMD	0.0143	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
17	IJK	0.0142	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
19	IVOG	0.0142	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
20	MDYG	0.0142	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
29	VXF	0.0142	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
69	FYT	0.0142	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
41	DES	0.0141	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
62	XMHQ	0.0141	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
63	RFV	0.0141	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
65	OMFS	0.0141	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
28	ISCG	0.0140	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
14	IWO	0.0139	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
22	VTWG	0.0139	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
66	RZV	0.0139	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
26	VBK	0.0138	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
35	FXD	0.0138	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
61	PSCI	0.0138	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
68	XSMO	0.0138	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
10	IWC	0.0136	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
39	RSPD	0.0135	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
51	SMMV	0.0135	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	FLAT
6	REGL	0.0134	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
37	FTXR	0.0133	First Trust Nasdaq Transportation ETF	Industrials	LONG
40	SMDV	0.0133	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
67	PSCD	0.0133	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
46	XTN	0.0131	State Street SPDR S&P Transportation ETF	Industrials	FLAT
49	XSLV	0.0131	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT

1	XRT	0.0128	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
48	XMLV	0.0127	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
36	PEJ	0.0126	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
38	IYT	0.0126	iShares U.S. Transportation ETF	Industrials	FLAT
2	JETS	0.0115	U.S. Global Jets ETF	Industrials	FLAT
3	REM	0.0114	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 3 – U.S. Large Cap Value

Members: 61 **Var explained:** 0.896 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.8%)
- Latest sell_cross: 2026-05-21
- Current 252d return: +22.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-21 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
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U.S. Large Cap Value	10	0.1697
U.S. Dividend Equity	8	0.1334
Industrials	7	0.1128
Materials	4	0.0618
U.S. Mid Cap Equity	3	0.0506
U.S. Multifactor Equity	3	0.0502
U.S. Value Factor Equity	3	0.0497
Utilities	3	0.0462
U.S. Fundamental Equity	2	0.0341
U.S. Equal Weight Equity	2	0.0341
U.S. Mid Cap Value	2	0.0340
U.S. Equity	2	0.0337
Consumer Discretionary	2	0.0268
Large Cap Blend Equities	1	0.0170
U.S. Size Factor Equity	1	0.0169
U.S. Revenue Weighted Equity	1	0.0168
Value Equity	1	0.0167
U.S. Mega Cap Value	1	0.0166
U.S. Buyback Equity	1	0.0165
U.S. Large Cap Equity	1	0.0164
U.S. ESG Equity	1	0.0162
U.S. Moat Equity	1	0.0159
Communication Services	1	0.0137

Replication basket – VTV (cor=0.97) basket lead at lag 1: -0.000

ticker	rep_wt	name	category	etf_signal
VTV	1.098	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	RSP	0.0172	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
26	IWD	0.0172	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
15	PRF	0.0171	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
18	CFA	0.0171	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
36	VONV	0.0171	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
42	SCHV	0.0171	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
48	IWS	0.0171	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
6	FEX	0.0170	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
22	FNDX	0.0170	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG

29	JPUS	0.0170	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
39	ILCV	0.0170	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
41	IUSV	0.0170	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
60	JPME	0.0170	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
23	EQAL	0.0169	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
27	IWR	0.0169	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
28	IVE	0.0169	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
30	SIZE	0.0169	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
34	VOOV	0.0169	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
40	IMCB	0.0169	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	SPYV	0.0169	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
45	VTV	0.0169	Vanguard Value ETF	U.S. Large Cap Value	LONG
49	VOE	0.0169	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
61	JVAL	0.0169	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
16	RWL	0.0168	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
20	NULV	0.0168	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
25	ONEV	0.0168	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
44	VO	0.0168	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
52	DTD	0.0168	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
54	DGRO	0.0168	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
12	IYJ	0.0167	iShares U.S. Industrials ETF	Industrials	FLAT
17	AIVL	0.0167	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	IWX	0.0167	iShares Russell Top 200 Value ETF	Value Equity	LONG
38	VYM	0.0167	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
50	PFM	0.0167	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
51	DLN	0.0167	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
55	VIG	0.0167	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
37	MGV	0.0166	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
4	FXR	0.0165	First Trust Industrials AlphaDEX	Industrials	LONG
5	PKW	0.0165	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	RDVY	0.0165	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
21	FDVV	0.0165	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
33	VIS	0.0164	Vanguard Industrials ETF	Industrials	FLAT
35	FIDU	0.0164	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
53	DIA	0.0164	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
1	XLI	0.0163	State Street Industrial Select Sector S...	Industrials	FLAT
31	VLUE	0.0162	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
56	ETHO	0.0162	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
19	CFO	0.0161	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
57	PAVE	0.0160	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
11	MOAT	0.0159	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
9	FIW	0.0158	First Trust Water ETF	Utilities	LONG
32	VAW	0.0158	Vanguard Materials ETF	Materials	FLAT
2	XLB	0.0156	State Street Materials Select Sector SP...	Materials	LONG
7	PHO	0.0156	Invesco Water Resources ETF	Utilities	FLAT

14	RSPM	0.0154	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
13	IYM	0.0150	iShares U.S. Basic Materials ETF	Materials	LONG
8	CGW	0.0147	Invesco S&P Global Water Index ETF	Utilities	FLAT
59	PKB	0.0146	Invesco Building & Construction ETF	Industrials	LONG
47	XHB	0.0140	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
58	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	FLAT
46	ITB	0.0128	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG

Cluster 4 – Developed ex-US Equity

Members: 61 Var explained: 0.898 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +24.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3158
Developed ex-US Equity	9	0.1527
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0660
Alternatives	4	0.0615
Eurozone Equity	3	0.0490
Commodity	1	0.0171
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
United Kingdom Equity	1	0.0158
Europe Financials	1	0.0158
Netherlands Equity	1	0.0157
Sweden Equity	1	0.0156
Spain Equity	1	0.0150
Austria Equity	1	0.0148
Belgium Equity	1	0.0146
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
VEA	0.967	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0172	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0172	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	LONG
40	GSIE	0.0171	Goldman Sachs ActiveBeta International ...	Commodity	LONG
6	IEV	0.0170	iShares Europe ETF	Europe Equity	FLAT
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	FLAT

19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
32	INTF	0.0170	iShares International Equity Factor ETF	International Equity	LONG
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	FLAT
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
49	VGK	0.0170	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	FLAT
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
38	VYMI	0.0168	Vanguard International High Dividend Yi...	International Dividend E...	LONG
14	EZU	0.0167	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	FLAT
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	FLAT
26	JPIN	0.0167	JPMorgan Diversified Return Internation...	International Equity	FLAT
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
34	IVLU	0.0166	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	FLAT
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	FLAT
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	FLAT
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	LONG
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
42	HDEF	0.0158	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
58	EWN	0.0157	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	FLAT

8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0150	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EW0	0.0148	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWK	0.0146	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 5 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-29 (10 days, -0.3%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +5.5%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
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Fixed Income	19	0.3283
Investment Grade Corporate Bond	13	0.2302
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0890
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0314
Target Maturity Bond	2	0.0307
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.007

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT

3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
52	SPMB	0.0179	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	LONG
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	LONG
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0156	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 6 – U.S. Large Cap Growth

Members: 56 Var explained: 0.838 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +33.6%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +34.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Technology	30	0.5299
U.S. Large Cap Growth	11	0.2094
Growth Equity	2	0.0375
Convertible Bond	2	0.0342
Financials	2	0.0332
U.S. Equity	1	0.0192
U.S. Mega Cap Growth	1	0.0189
U.S. Equal Weight Equity	1	0.0188
Options Income	1	0.0171
U.S. Small Cap Technology	1	0.0171
IPO Equity	1	0.0171
Trend-Following Equity	1	0.0167
Consumer Discretionary	1	0.0163
U.S. Trend-Following Equity	1	0.0145

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.020

ticker	rep_wt	name	category	etf_signal
QQQ	1.036	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
12	QQQ	0.0192	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
31	IGM	0.0192	iShares Expanded Tech Sector ETF	Technology	FLAT
36	ONEQ	0.0192	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
46	ILCG	0.0192	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
25	XNTK	0.0191	State Street SPDR NYSE Technology ETF	Technology	LONG
33	IYW	0.0191	iShares U.S. Technology ETF	Technology	FLAT
39	IWF	0.0191	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
41	VGT	0.0191	Vanguard Information Tech ETF	Technology	FLAT
43	FTEC	0.0191	Fidelity MSCI Information Technology In...	Technology	FLAT
50	VUG	0.0191	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
15	QTEC	0.0190	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
35	NULG	0.0190	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
44	VONG	0.0190	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
47	IUSG	0.0190	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
48	SCHG	0.0190	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
20	FXL	0.0189	First Trust Technology AlphaDEX	Technology	LONG
40	IVW	0.0189	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
42	VOOG	0.0189	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
45	MGK	0.0189	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
49	SPYG	0.0189	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
1	XLK	0.0188	State Street Technology Select Sector S...	Technology	FLAT
6	AIQ	0.0188	Global X Artificial Intelligence & Tech...	Technology	FLAT
14	QQEW	0.0188	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
32	IXN	0.0188	iShares Global Tech ETF	Technology	FLAT
38	IWY	0.0188	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
30	PWB	0.0186	Invesco Large Cap Growth ETF	Growth Equity	FLAT
17	TDIV	0.0179	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
24	FDN	0.0179	First Trust DJ Internet Index Fund	Technology	FLAT
56	QTUM	0.0178	Defiance Quantum ETF	Technology	LONG
23	PNQI	0.0177	Invesco Nasdaq Internet ETF	Technology	FLAT
3	SMH	0.0176	VanEck Semiconductor ETF	Technology	LONG
5	SKYY	0.0176	First Trust Cloud Computing ETF	Technology	LONG
2	SOXX	0.0175	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
55	PTF	0.0174	Invesco Dorsey Wright Technology Moment...	Technology	LONG
11	ARKQ	0.0173	ARK Autonomous Technology & Robotics ETF	Technology	FLAT
18	XSD	0.0173	State Street SPDR S&P Semiconductor ETF	Technology	LONG

34	CWB	0.0173	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
4	IGV	0.0172	iShares Expanded Tech-Software Sector E...	Technology	FLAT
13	NXTG	0.0172	First Trust Indxx NextG ETF	Technology	LONG
16	PSI	0.0171	Invesco Semiconductors ETF	Technology	LONG
22	IPO	0.0171	Renaissance IPO ETF	IPO Equity	LONG
29	QYLD	0.0171	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
52	PSCT	0.0171	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
10	ARKW	0.0170	ARK Next Generation Internet ETF	Technology	FLAT
27	FINX	0.0170	Global X FinTech ETF	Financials	FLAT
54	FTXL	0.0170	First Trust Nasdaq Semiconductor ETF	Technology	LONG
37	ICVT	0.0169	iShares Convertible Bond ETF	Convertible Bond	FLAT
9	ARKK	0.0167	ARK Innovation ETF	Technology	FLAT
28	PTNQ	0.0167	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
21	CIBR	0.0165	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
8	HACK	0.0163	Amplify Cybersecurity ETF	Technology	LONG
19	IBUY	0.0163	Amplify Online Retail ETF	Consumer Discretionary	LONG
26	IPAY	0.0162	Amplify Digital Payments ETF	Financials	FLAT
7	BLOK	0.0154	Amplify Blockchain Technology ETF	Technology	FLAT
51	PTLC	0.0145	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT
53	BLCN	0.0136	Siren NexGen Economy ETF	Technology	LONG

Cluster 7 – Emerging Markets Equity

Members: 25 **Var explained:** 0.874 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +37.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7725
Asia Equity	2	0.0813
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0360
Global Dividend Equity	1	0.0344
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.028

ticker	rep_wt	name	category	etf_signal
IEMG	0.925	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT

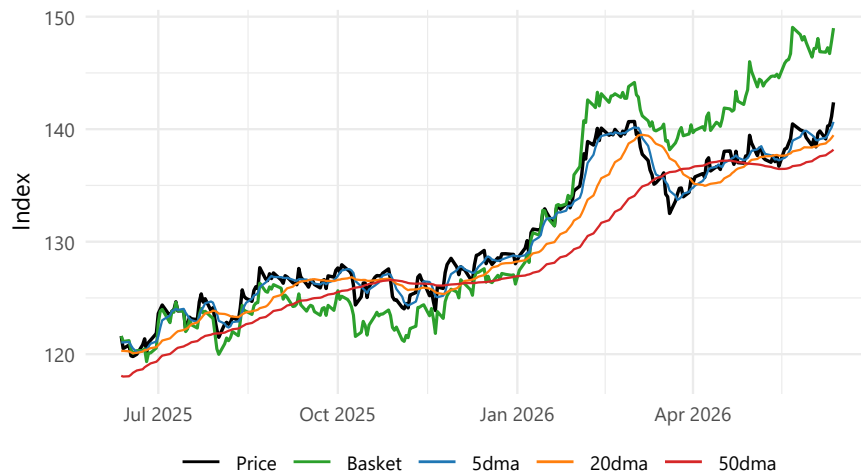
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0422	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0402	iShares Asia 50 ETF	Asia Equity	FLAT
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0391	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0387	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0360	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0348	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
24	SDIV	0.0344	Global X SuperDividend ETF	Global Dividend Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT

Cluster 8 – U.S. Dividend Equity

Members: 25 **Var explained:** 0.871 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +17.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-20)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5281
U.S. Low Volatility Equity	4	0.1561
Value Equity	3	0.1201
Dividend Equity	2	0.0792
U.S. Defensive Equity	1	0.0403
U.S. Quality / Factor Equity	1	0.0390
Global Equity	1	0.0371

Replication basket – SCHD (cor=0.96) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
SCHD	0.894	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
7 SDY	0.0418	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG

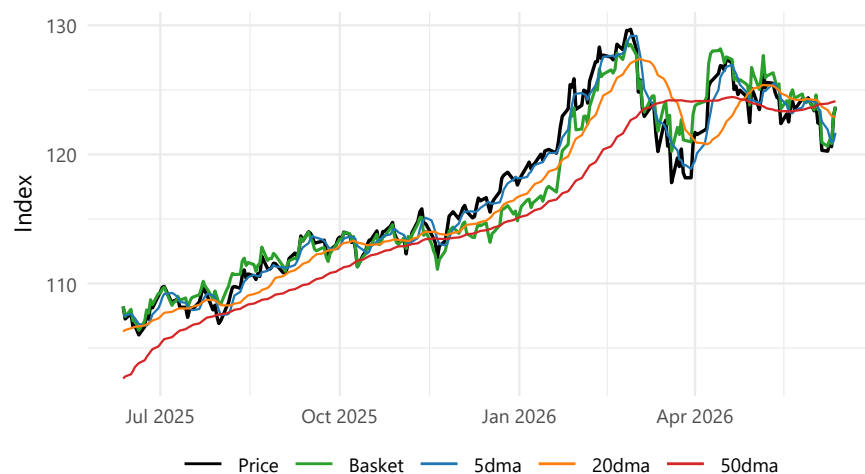
15	DVY	0.0417	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
2	FVD	0.0415	First Trust VL Dividend	U.S. Dividend Equity	LONG
10	SCHD	0.0413	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
14	SDOG	0.0412	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
20	SPYD	0.0412	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
5	DHS	0.0411	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
11	FTA	0.0411	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
19	ONEY	0.0410	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
18	SPHD	0.0408	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
6	CDC	0.0403	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
4	FDL	0.0402	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
25	LGLV	0.0400	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
3	PWV	0.0399	Invesco Large Cap Value ETF	Value Equity	LONG
8	LVHD	0.0399	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
9	HDV	0.0394	iShares Core High Dividend ETF	U.S. Dividend Equity	FLAT
16	RDIV	0.0394	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
12	PEY	0.0393	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
13	DIV	0.0391	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
17	RPV	0.0391	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
21	FTCS	0.0390	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
22	OUSA	0.0384	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
24	USMV	0.0382	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
23	SPLV	0.0380	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
1	ACWV	0.0371	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG

Cluster 9 – Asia-Pacific Equity

Members: 23 **Var explained:** 0.686 **Signal:** LONG since 2026-05-22 (14 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +17.2%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +14.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-22 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0978
International Dividend Equity	2	0.0959
International Equity	2	0.0957
Global Equity	2	0.0919
Real Estate	2	0.0917
Emerging Markets Fixed Income	2	0.0838
Currency	2	0.0812
Australia Equity	1	0.0486
Canada Equity	1	0.0461
Switzerland Equity	1	0.0431
South Africa Equity	1	0.0411
Latin America Equity	1	0.0396
New Zealand Equity	1	0.0394
Mexico Equity	1	0.0374
Brazil Equity	1	0.0335
Chile Equity	1	0.0332

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
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EWZ	0.095	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.289	iShares Global Infrastructure ETF	Global Equity	FLAT
BBAX	0.453	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	EPP	0.0489	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
23	BBAX	0.0489	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
11	IDLV	0.0488	Invesco S&P International Developed Low...	International Equity	FLAT
4	EWA	0.0486	iShares MSCI Australia Index Fund	Australia Equity	FLAT
10	DWX	0.0482	State Street SPDR S&P International Div...	International Dividend E...	LONG
3	PID	0.0477	Invesco International Dividend Achiever...	International Dividend E...	FLAT
12	EFAV	0.0470	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
8	NFRA	0.0466	FlexShares STOXX Global Broad Infrastru...	Global Equity	FLAT
13	VNQI	0.0464	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
6	EWC	0.0461	iShares MSCI Canada Index Fund	Canada Equity	LONG
2	RWX	0.0453	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
9	IGF	0.0453	iShares Global Infrastructure ETF	Global Equity	FLAT
5	EWL	0.0431	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
19	FXA	0.0431	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
14	EBND	0.0420	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
15	EMLC	0.0418	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
21	EZA	0.0411	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
18	ILF	0.0396	iShares Latin America 40 ETF	Latin America Equity	FLAT
20	ENZL	0.0394	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
22	FXC	0.0381	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
16	EWX	0.0374	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0335	iShares MSCI Brazil ETF	Brazil Equity	FLAT
17	ECH	0.0332	iShares MSCI Chile ETF	Chile Equity	FLAT

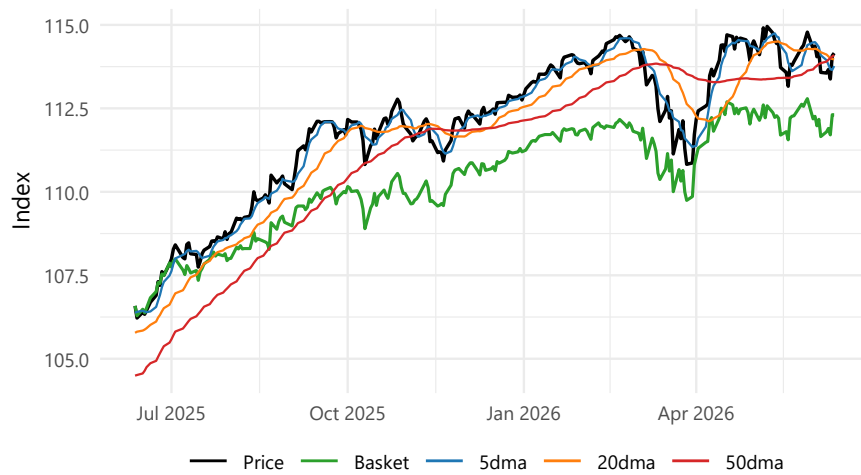
Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +7.1% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5155
Preferred Stock	7	0.3062
Emerging Markets Fixed Income	4	0.1784

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.048

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT

18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT
14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFXF	0.0462	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.762 **Signal:** LONG since 2026-05-06 (26 days, 3.2%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +16.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
VHT	1.101	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	LONG

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0571	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0543	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0509	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0471	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – Materials

Members: 17 **Var explained:** 0.692 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +48.9%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +38.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5582
MLP / Energy Infrastructure	4	0.2190
Energy	1	0.0642
Consumer Staples	1	0.0598
Norway Equity	1	0.0557
Industrial Metals	1	0.0431

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.045

ticker	rep_wt	name	category	etf_signal
AMLP	0.142	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.093	Global X Uranium ETF	Materials	FLAT
GUNR	0.776	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0689	State Street SPDR S&P Global Natural Re...	Materials	FLAT
10	GUNR	0.0686	FlexShares Global Upstream Natural Reso...	Materials	FLAT
17	NANR	0.0665	State Street SPDR S&P North American Na...	Materials	FLAT
16	IGE	0.0642	iShares North American Natural Resource...	Energy	FLAT
9	MXI	0.0634	iShares Global Materials ETF	Materials	FLAT
12	PICK	0.0631	iShares MSCI Global Select Metals & Min...	Materials	FLAT
3	XME	0.0613	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
13	SLX	0.0605	VanEck Steel ETF	Materials	FLAT
5	MOO	0.0598	VanEck Agribusiness ETF	Consumer Staples	FLAT
2	COPX	0.0590	Global X Copper Miners ETF	Materials	FLAT
8	MLPX	0.0569	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
14	NORW	0.0557	Global X MSCI Norway ETF	Norway Equity	FLAT
4	EMLP	0.0553	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	FLAT
6	AMLP	0.0541	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
7	MLPA	0.0528	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0468	Global X Uranium ETF	Materials	FLAT
15	DBB	0.0431	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 13 – China Equity**Members:** 15 **Var explained:** 0.802 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +7.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0737	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0595	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	FLAT
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0586	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.877 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +13.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.938
Global Equity	1	0.062

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0754	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0754	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0752	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0751	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0747	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0740	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	FLAT
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0699	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0638	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0620	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +14.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.000

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

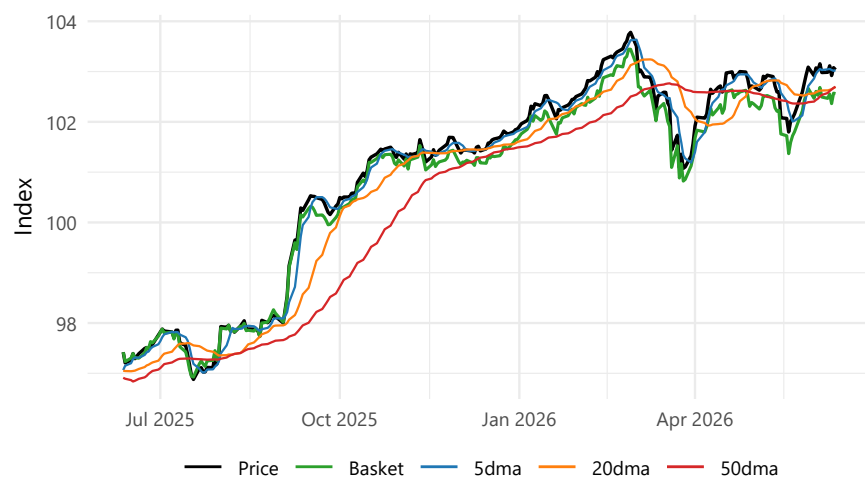
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0715	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (11 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.179

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.860 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +36.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4158
Gold	3	0.2501
Precious Metals	2	0.1725
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
GLD	0.037	SPDR Gold Shares	Gold	FLAT
IAU	0.733	iShares Gold Trust Shares	Gold	FLAT
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0832	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.745 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +8.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-29 more recent than last effective buy 2026-05-22)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5732
Utilities	4	0.3521
U.S. Small Cap Consumer Staples	1	0.0747

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0949	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

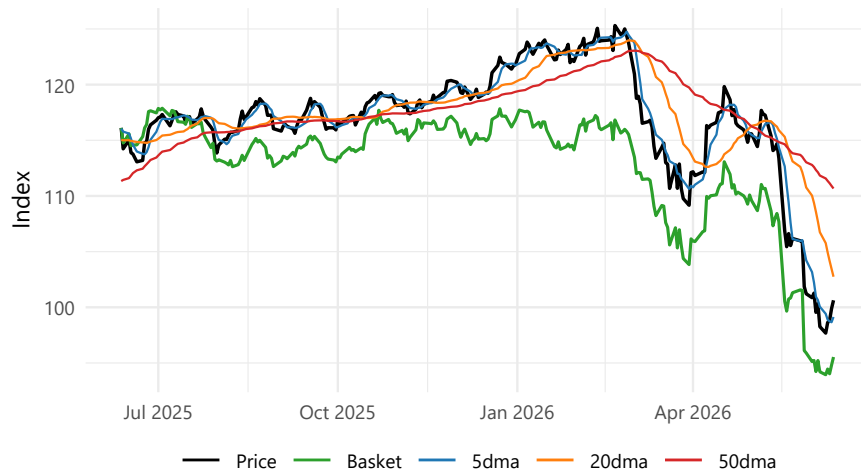
3	KXI	0.0946	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0890	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0878	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0878	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0747	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – India Equity

Members: 11 **Var explained:** 0.369 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.4%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -13.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4649
Malaysia Equity	1	0.1129
Indonesia Equity	1	0.1045
Philippines Equity	1	0.1009
Vietnam Equity	1	0.0743
Turkey Equity	1	0.0603
U.S. Dividend Equity	1	0.0546
Platinum	1	0.0230
Natural Gas	1	0.0047

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.658	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.028	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

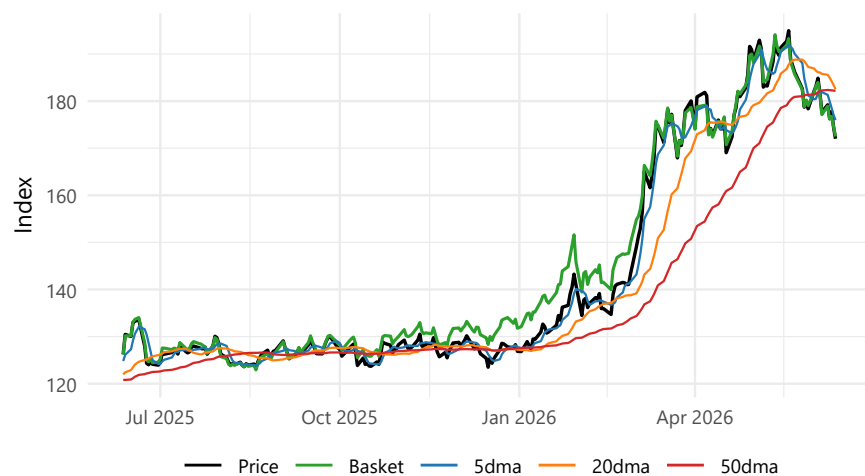
	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1563	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1547	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1539	WisdomTree India Earnings Fund	India Equity	FLAT
9	EWM	0.1129	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
7	EIDO	0.1045	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
8	EPHE	0.1009	iShares MSCI Philippines ETF	Philippines Equity	FLAT
11	VNM	0.0743	VanEck Vietnam ETF	Vietnam Equity	FLAT
10	TUR	0.0603	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0546	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0230	abrdn Physical Platinum Shares ETF	Platinum	FLAT
6	UNG	0.0047	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – Broad Commodities

Members: 10 **Var explained:** 0.841 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.8%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +36.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-27 more recent than last effective buy 2026-01-08)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6371
Oil	2	0.2056
Energy Commodities	1	0.1053
Agricultural Commodities	1	0.0520

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.073

ticker	rep_wt	name	category	etf_signal
PDBC	1.129	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1091	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1091	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1091	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1089	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1053	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.1034	United States Oil Fund	Oil	LONG
5	DBO	0.1022	Invesco DB Oil Fund	Oil	FLAT
7	DJP	0.1006	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.1003	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0520	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.906 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.263	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.762	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

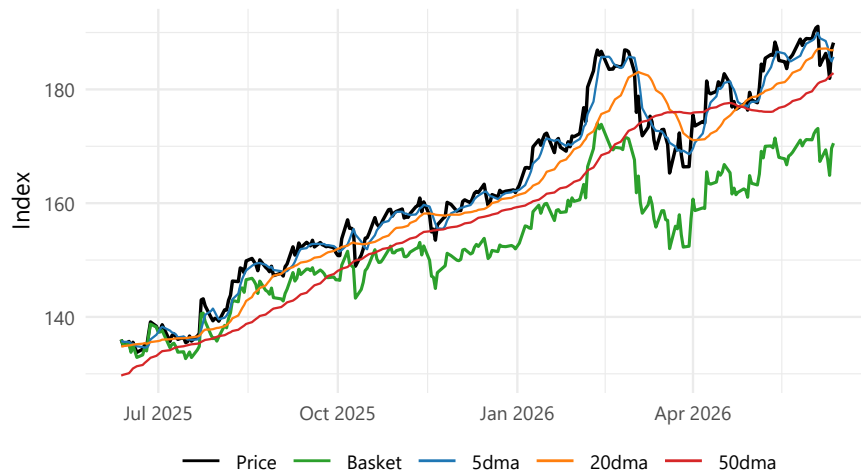
	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1034	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1033	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDDT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0999	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0994	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +38.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7802
Asia-Pacific Equity	2	0.2198

Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
EWJ	0.909	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1117	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1116	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1108	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT

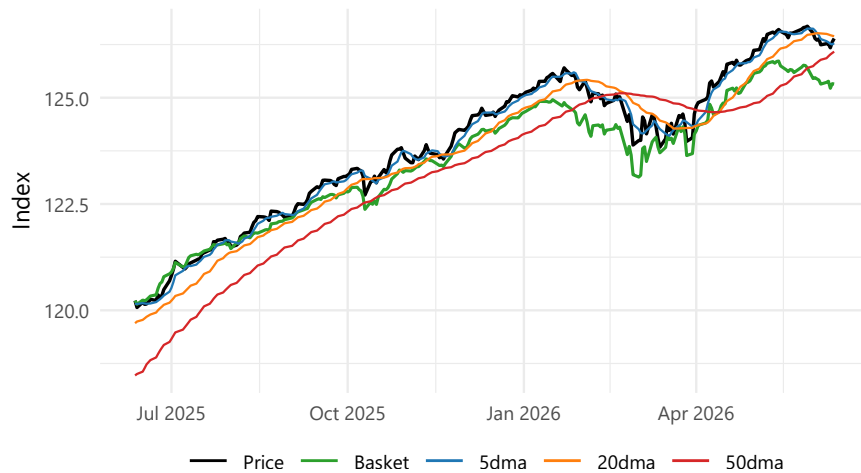
2	DXJ	0.1096	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1085	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.558 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-04
- Current 252d return: +5.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-04 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3628
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income	3	0.3127
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Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.211	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.308	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

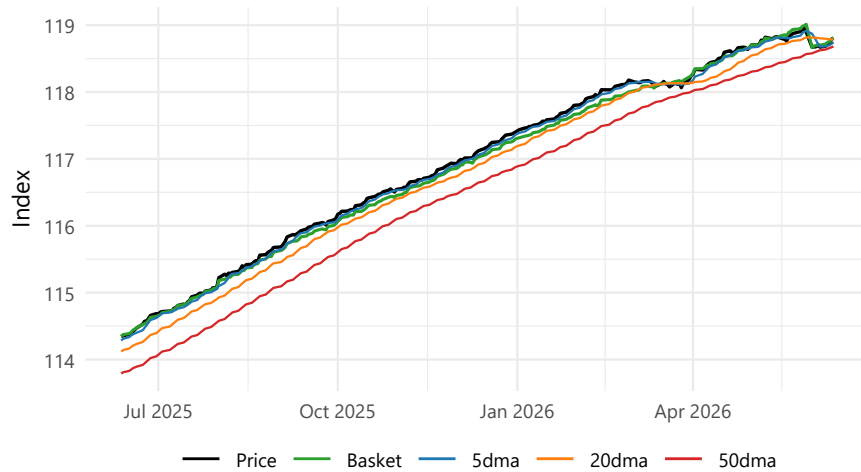
	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT
3	BKLN	0.1211	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1167	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1037	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1009	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.453 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.096

ticker	rep_wt	name	category	etf_signal
SHV	0.295	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.178	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.122	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.410	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1280	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1274	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1265	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1203	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT

2	MINT	0.1100	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1098	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0989	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.0948	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0843	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (10 days, -1.0%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -1.5% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409

International Fixed Income	2	0.3591
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Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.026

ticker	rep_wt	name	category	etf_signal
BWX	0.75	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1648	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXV	0.1450	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse

Members: 2 **Var explained:** 0.626 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-04-08
- Current 252d return: -38.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)



Category breakdown

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.021

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
RSPH	cluster_11	Healthcare	2026-05-21	15	4.4%	0.00	1.00
XLV	cluster_11	Healthcare	2026-05-21	15	3.8%	0.00	1.00
FHLC	cluster_11	Healthcare	2026-05-19	17	4.6%	0.00	1.00
VHT	cluster_11	Healthcare	2026-05-19	17	4.5%	0.00	1.00
IYH	cluster_11	Healthcare	2026-05-20	16	4.2%	0.00	1.00
PSCH	cluster_11	Healthcare	2026-04-13	43	11.5%	0.00	1.00
FXH	cluster_11	Healthcare	2026-04-10	44	5.1%	0.00	1.00
IXJ	cluster_11	Healthcare	2026-05-22	14	1.5%	0.00	1.00
PPH	cluster_11	Healthcare	2026-05-21	15	2.6%	0.00	1.00
IHF	cluster_11	Healthcare	2026-05-08	24	7.0%	0.01	0.99
XHE	cluster_11	Healthcare	2026-06-09	3	-1.5%	0.00	0.99
IBB	cluster_11	Healthcare	2026-05-28	11	-0.9%	0.43	-0.35
IAT	cluster_14	Financials	2026-05-20	16	7.9%	0.00	1.00
XLF	cluster_14	Financials	2026-04-13	43	3.3%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	43	3.3%	0.00	1.00
FXO	cluster_14	Financials	2026-06-05	5	2.7%	0.00	1.00
KIE	cluster_14	Financials	2026-06-10	2	0.6%	0.04	0.83
IAI	cluster_14	Financials	2026-04-10	44	8.4%	0.26	0.72
RPV	cluster_8	U.S. Dividend Equity	2026-04-20	38	6.5%	0.00	1.00
PWV	cluster_8	U.S. Dividend Equity	2026-04-02	49	10.0%	0.00	1.00
SPYD	cluster_8	U.S. Dividend Equity	2026-04-21	37	6.3%	0.00	1.00
FTA	cluster_8	U.S. Dividend Equity	2026-05-21	15	3.5%	0.00	1.00
ONEY	cluster_8	U.S. Dividend Equity	2026-04-10	44	7.5%	0.00	1.00
SDY	cluster_8	U.S. Dividend Equity	2026-05-20	16	3.1%	0.00	1.00
SDOG	cluster_8	U.S. Dividend Equity	2026-04-07	47	7.9%	0.00	1.00
SPHD	cluster_8	U.S. Dividend Equity	2026-06-08	4	3.3%	0.00	1.00
RDIV	cluster_8	U.S. Dividend Equity	2026-04-21	37	7.4%	0.00	1.00
LVHD	cluster_8	U.S. Dividend Equity	2026-06-10	2	0.6%	0.00	0.99
FDL	cluster_8	U.S. Dividend Equity	2026-06-05	5	1.6%	0.03	0.97
FVD	cluster_8	U.S. Dividend Equity	2026-06-10	2	1.2%	0.01	0.94
FTCS	cluster_8	U.S. Dividend Equity	2026-06-10	2	1.0%	0.09	0.85
SPLV	cluster_8	U.S. Dividend Equity	2026-06-11	1	0.9%	0.04	0.74
LGLV	cluster_8	U.S. Dividend Equity	2026-06-12	0	0.0%	0.01	0.72
SCHD	cluster_8	U.S. Dividend Equity	2026-04-21	37	5.9%	0.39	0.61
OUSA	cluster_8	U.S. Dividend Equity	2026-04-20	38	1.6%	0.71	0.27
ACWV	cluster_8	U.S. Dividend Equity	2026-06-12	0	0.0%	0.92	-0.34
PEJ	cluster_2	U.S. Mid Cap Equity	2026-05-26	13	5.8%	0.00	1.00

PSCD	cluster_2	U.S. Mid Cap Equity	2026-05-27	12	5.3%	0.00	1.00
IJS	cluster_2	U.S. Mid Cap Equity	2026-04-10	44	10.2%	0.00	1.00
XRT	cluster_2	U.S. Mid Cap Equity	2026-05-28	11	2.5%	0.00	0.99
ISMD	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	14.4%	0.01	0.99
FXD	cluster_2	U.S. Mid Cap Equity	2026-05-27	12	2.7%	0.00	0.99
FTXR	cluster_2	U.S. Mid Cap Equity	2026-04-17	39	9.1%	0.03	0.97
RSPD	cluster_2	U.S. Mid Cap Equity	2026-06-11	1	0.5%	0.00	0.95
SCHA	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	13.0%	0.22	0.77
SCHM	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	10.9%	0.24	0.76
SDVY	cluster_2	U.S. Mid Cap Equity	2026-06-08	4	2.4%	0.01	0.60
REGL	cluster_2	U.S. Mid Cap Equity	2026-06-12	0	0.0%	0.01	0.48
IVOG	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	8.9%	0.51	0.44
XMLV	cluster_2	U.S. Mid Cap Equity	2026-06-12	0	0.0%	0.09	0.44
MDYG	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	8.9%	0.51	0.43
IJK	cluster_2	U.S. Mid Cap Equity	2026-04-09	45	8.8%	0.52	0.43
VTWG	cluster_2	U.S. Mid Cap Equity	2026-04-13	43	12.3%	0.64	0.26
IWO	cluster_2	U.S. Mid Cap Equity	2026-04-13	43	12.2%	0.64	0.26
NURE	cluster_15	Real Estate	2026-04-13	43	12.7%	0.00	1.00
INDS	cluster_15	Real Estate	2026-06-08	4	3.8%	0.04	0.87
RWO	cluster_15	Real Estate	2026-06-09	3	1.7%	0.15	0.64
SHM	cluster_16	Municipal Bond	2026-05-28	11	0.0%	0.01	0.98
FMB	cluster_16	Municipal Bond	2026-05-29	10	0.2%	0.01	0.95
HYD	cluster_16	Municipal Bond	2026-05-22	14	1.2%	0.03	0.94
PZA	cluster_16	Municipal Bond	2026-05-29	10	0.3%	0.02	0.90
SUB	cluster_16	Municipal Bond	2026-05-27	12	0.0%	0.03	0.90
CMF	cluster_16	Municipal Bond	2026-05-28	11	0.1%	0.02	0.88
TFI	cluster_16	Municipal Bond	2026-05-28	11	0.0%	0.02	0.88
VTEB	cluster_16	Municipal Bond	2026-05-28	11	0.0%	0.03	0.88
NYF	cluster_16	Municipal Bond	2026-05-28	11	0.1%	0.03	0.87
HYMB	cluster_16	Municipal Bond	2026-05-26	13	0.6%	0.06	0.87
ITM	cluster_16	Municipal Bond	2026-05-28	11	-0.1%	0.10	0.72
MUB	cluster_16	Municipal Bond	2026-05-28	11	0.0%	0.07	0.72
MUNI	cluster_16	Municipal Bond	2026-05-28	11	-0.1%	0.08	0.64
PKW	cluster_3	U.S. Large Cap Value	2026-05-20	16	2.9%	0.00	1.00
MGV	cluster_3	U.S. Large Cap Value	2026-04-13	43	8.4%	0.00	1.00
VTV	cluster_3	U.S. Large Cap Value	2026-04-13	43	7.1%	0.00	1.00
DGRO	cluster_3	U.S. Large Cap Value	2026-04-13	43	5.0%	0.00	1.00
XHB	cluster_3	U.S. Large Cap Value	2026-05-29	10	4.8%	0.00	0.99
RWL	cluster_3	U.S. Large Cap Value	2026-04-10	44	8.9%	0.01	0.99
SCHV	cluster_3	U.S. Large Cap Value	2026-04-10	44	9.5%	0.01	0.99
ITB	cluster_3	U.S. Large Cap Value	2026-06-09	3	0.7%	0.00	0.99
RSP	cluster_3	U.S. Large Cap Value	2026-04-14	42	6.0%	0.04	0.96
IMCB	cluster_3	U.S. Large Cap Value	2026-04-10	44	9.8%	0.12	0.88
FEX	cluster_3	U.S. Large Cap Value	2026-04-09	45	8.3%	0.15	0.85
IWR	cluster_3	U.S. Large Cap Value	2026-04-10	44	8.0%	0.20	0.80

AIVL	cluster_3	U.S. Large Cap Value	2026-04-10	44	6.5%	0.20	0.80
VONV	cluster_3	U.S. Large Cap Value	2026-04-10	44	9.2%	0.21	0.79
VO	cluster_3	U.S. Large Cap Value	2026-04-13	43	6.5%	0.21	0.79
SIZE	cluster_3	U.S. Large Cap Value	2026-04-13	43	6.4%	0.22	0.78
IWD	cluster_3	U.S. Large Cap Value	2026-04-10	44	9.1%	0.23	0.77
JVAL	cluster_3	U.S. Large Cap Value	2026-04-10	44	14.4%	0.25	0.75
DIA	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.7%	0.27	0.73
VIG	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.3%	0.27	0.73
IWX	cluster_3	U.S. Large Cap Value	2026-04-10	44	9.7%	0.34	0.66
VYM	cluster_3	U.S. Large Cap Value	2026-04-10	44	5.4%	0.37	0.63
FDVV	cluster_3	U.S. Large Cap Value	2026-04-14	42	6.3%	0.38	0.62
PFM	cluster_3	U.S. Large Cap Value	2026-04-14	42	4.7%	0.43	0.57
FIW	cluster_3	U.S. Large Cap Value	2026-05-08	24	-0.1%	0.00	0.55
DTD	cluster_3	U.S. Large Cap Value	2026-04-13	43	5.3%	0.46	0.54
VLUE	cluster_3	U.S. Large Cap Value	2026-04-09	45	29.8%	0.50	0.50
FNDX	cluster_3	U.S. Large Cap Value	2026-04-09	45	8.7%	0.56	0.44
PRF	cluster_3	U.S. Large Cap Value	2026-04-09	45	9.3%	0.65	0.35
FXR	cluster_3	U.S. Large Cap Value	2026-06-12	0	0.0%	0.15	0.30
DLN	cluster_3	U.S. Large Cap Value	2026-04-13	43	5.4%	0.71	0.29
IUSV	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.1%	0.73	0.27
IVE	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.1%	0.86	0.14
SPYV	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.1%	0.86	0.13
VOOV	cluster_3	U.S. Large Cap Value	2026-04-14	42	5.2%	0.86	0.13
ILCV	cluster_3	U.S. Large Cap Value	2026-04-13	43	6.0%	0.96	0.04
NULV	cluster_3	U.S. Large Cap Value	2026-04-10	44	8.0%	0.96	0.03
PKB	cluster_3	U.S. Large Cap Value	2026-05-28	11	0.3%	0.22	-0.13
XLB	cluster_3	U.S. Large Cap Value	2026-05-28	11	1.6%	0.52	-0.46
RSPM	cluster_3	U.S. Large Cap Value	2026-05-29	10	1.9%	0.52	-0.48
IYM	cluster_3	U.S. Large Cap Value	2026-05-26	13	1.4%	0.65	-0.48
PSCC	cluster_18	Consumer Staples	2026-06-10	2	1.8%	0.00	1.00
IYK	cluster_18	Consumer Staples	2026-06-09	3	1.9%	0.09	0.90
XLP	cluster_18	Consumer Staples	2026-06-10	2	0.4%	0.24	0.69
FSTA	cluster_18	Consumer Staples	2026-06-11	1	0.7%	0.25	0.43
VDC	cluster_18	Consumer Staples	2026-06-11	1	0.7%	0.26	0.40
KXI	cluster_18	Consumer Staples	2026-06-11	1	0.4%	0.36	0.37
UUP	cluster_26	Leveraged / Inverse	2026-05-18	18	0.9%	0.00	1.00
IPO	cluster_6	U.S. Large Cap Growth	2026-04-13	43	20.6%	0.36	0.64
SOXX	cluster_6	U.S. Large Cap Growth	2026-04-08	46	61.0%	0.39	0.61
FXL	cluster_6	U.S. Large Cap Growth	2026-04-09	45	26.6%	0.47	0.53
QTUM	cluster_6	U.S. Large Cap Growth	2026-04-10	44	38.2%	0.59	0.41
IBUY	cluster_6	U.S. Large Cap Growth	2026-05-27	12	-0.4%	0.09	0.40
FTXL	cluster_6	U.S. Large Cap Growth	2026-04-07	47	72.6%	0.61	0.39
XNTK	cluster_6	U.S. Large Cap Growth	2026-04-10	44	33.5%	0.62	0.38
PSI	cluster_6	U.S. Large Cap Growth	2026-04-07	47	68.7%	0.63	0.37
SMH	cluster_6	U.S. Large Cap Growth	2026-04-08	46	46.6%	0.63	0.37

HACK	cluster_6	U.S. Large Cap Growth	2026-04-17	39	22.2%	0.71	0.29
QQEW	cluster_6	U.S. Large Cap Growth	2026-04-16	40	13.8%	0.73	0.27
PSCT	cluster_6	U.S. Large Cap Growth	2026-04-07	47	39.4%	0.74	0.26
QTEC	cluster_6	U.S. Large Cap Growth	2026-04-10	44	41.3%	0.79	0.21
NXTG	cluster_6	U.S. Large Cap Growth	2026-04-08	46	29.6%	0.79	0.21
SKYY	cluster_6	U.S. Large Cap Growth	2026-04-17	39	14.4%	0.76	0.19
CIBR	cluster_6	U.S. Large Cap Growth	2026-04-17	39	28.4%	0.82	0.18
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	47	39.4%	0.83	0.10
XSD	cluster_6	U.S. Large Cap Growth	2026-04-09	45	66.3%	0.94	0.06
BLCN	cluster_6	U.S. Large Cap Growth	2026-04-15	41	11.0%	0.88	0.03
IAGG	cluster_5	Aggregate Fixed Income	2026-05-22	14	0.6%	0.12	0.85
EDV	cluster_5	Aggregate Fixed Income	2026-06-12	0	0.0%	0.04	0.72
BNDX	cluster_5	Aggregate Fixed Income	2026-05-26	13	0.1%	0.37	0.33
SPLB	cluster_5	Aggregate Fixed Income	2026-06-12	0	0.0%	0.25	0.27
BLV	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.2%	0.19	0.22
IGLB	cluster_5	Aggregate Fixed Income	2026-06-12	0	0.0%	0.27	0.21
VCLT	cluster_5	Aggregate Fixed Income	2026-06-12	0	0.0%	0.29	0.18
GTO	cluster_5	Aggregate Fixed Income	2026-06-01	9	0.0%	0.24	-0.06
AGGY	cluster_5	Aggregate Fixed Income	2026-06-01	9	0.1%	0.24	-0.10
LQD	cluster_5	Aggregate Fixed Income	2026-05-28	11	-0.2%	0.38	-0.16
SPMB	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.32	-0.19
FLCO	cluster_5	Aggregate Fixed Income	2026-05-27	12	-0.1%	0.42	-0.22
AGG	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.38	-0.28
BND	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.38	-0.29
IUSB	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.40	-0.29
MBB	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.4%	0.42	-0.31
SCHZ	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.40	-0.31
USIG	cluster_5	Aggregate Fixed Income	2026-05-27	12	-0.1%	0.48	-0.32
QLTA	cluster_5	Aggregate Fixed Income	2026-05-28	11	-0.3%	0.46	-0.34
BOND	cluster_5	Aggregate Fixed Income	2026-05-29	10	-0.3%	0.41	-0.34
DBJP	cluster_22	Japan Equity	2026-04-08	46	5.3%	0.87	0.11
HEWJ	cluster_22	Japan Equity	2026-04-08	46	5.4%	0.90	0.07
IPAC	cluster_22	Japan Equity	2026-06-12	0	0.0%	1.00	-0.87
SPHQ	cluster_1	U.S. Large Cap Equity	2026-04-10	44	10.9%	0.10	0.90
ITA	cluster_1	U.S. Large Cap Equity	2026-05-26	13	1.9%	0.15	0.84
IMCG	cluster_1	U.S. Large Cap Equity	2026-04-09	45	13.6%	0.41	0.59
FV	cluster_1	U.S. Large Cap Equity	2026-04-14	42	14.8%	0.48	0.52
MTUM	cluster_1	U.S. Large Cap Equity	2026-04-08	46	24.9%	0.49	0.51
PDP	cluster_1	U.S. Large Cap Equity	2026-04-08	46	12.7%	0.53	0.44
QUAL	cluster_1	U.S. Large Cap Equity	2026-04-14	42	6.7%	0.58	0.42
XAR	cluster_1	U.S. Large Cap Equity	2026-05-22	14	3.2%	0.51	0.41
PPA	cluster_1	U.S. Large Cap Equity	2026-05-27	12	-0.2%	0.40	0.40
SPHB	cluster_1	U.S. Large Cap Equity	2026-04-10	44	21.7%	0.59	0.38
VFMO	cluster_1	U.S. Large Cap Equity	2026-04-08	46	13.8%	0.57	0.37
RPG	cluster_1	U.S. Large Cap Equity	2026-04-09	45	19.3%	0.65	0.34

VOT	cluster_1	U.S. Large Cap Equity	2026-04-14	42	7.4%	0.63	0.31
FTC	cluster_1	U.S. Large Cap Equity	2026-04-10	44	13.0%	0.67	0.30
IWP	cluster_1	U.S. Large Cap Equity	2026-04-15	41	4.1%	0.68	0.23
XMMO	cluster_1	U.S. Large Cap Equity	2026-04-06	48	15.0%	0.79	-0.04
EWT	cluster_7	Emerging Markets Equity	2026-04-08	46	34.4%	0.50	0.50
DEM	cluster_7	Emerging Markets Equity	2026-04-09	45	8.9%	0.69	0.24
EWN	cluster_4	Developed ex-US Equity	2026-04-10	44	11.7%	0.03	0.97
IHDG	cluster_4	Developed ex-US Equity	2026-05-06	26	2.3%	0.09	0.91
EWI	cluster_4	Developed ex-US Equity	2026-04-08	46	7.2%	0.08	0.90
EWO	cluster_4	Developed ex-US Equity	2026-04-10	44	9.4%	0.21	0.78
IQLT	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.59	-0.20
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.72	-0.36
INTF	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.85	-0.59
VYMI	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.91	-0.65
GSIE	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.89	-0.69
RODM	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.93	-0.77
FEP	cluster_4	Developed ex-US Equity	2026-04-09	45	3.1%	0.97	-0.79
EFV	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	0.98	-0.87
TLTD	cluster_4	Developed ex-US Equity	2026-06-12	0	0.0%	1.00	-0.92
PCY	cluster_10	High Yield Bond	2026-05-28	11	0.9%	0.10	0.54
VWOB	cluster_10	High Yield Bond	2026-05-26	13	0.8%	0.25	0.26
HYEM	cluster_10	High Yield Bond	2026-04-13	43	1.1%	0.72	-0.06
FALN	cluster_10	High Yield Bond	2026-06-12	0	0.0%	0.43	-0.18
FPE	cluster_10	High Yield Bond	2026-05-27	12	-0.1%	0.93	-0.93
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-06-01	9	-0.4%	0.98	-0.98
ENZL	cluster_9	Asia-Pacific Equity	2026-05-22	14	3.3%	0.51	0.21
DWX	cluster_9	Asia-Pacific Equity	2026-06-12	0	0.0%	0.43	0.17
EWL	cluster_9	Asia-Pacific Equity	2026-06-11	1	-0.3%	0.79	-0.25
EWC	cluster_9	Asia-Pacific Equity	2026-04-09	45	4.1%	0.77	-0.26
TDTF	cluster_21	Inflation-Protected Fix...	2026-05-28	11	-1.5%	1.00	-1.00
MLPX	cluster_12	Materials	2026-06-08	4	1.1%	0.90	-0.36
MLPA	cluster_12	Materials	2026-06-08	4	-1.3%	0.96	-0.73
DBE	cluster_20	Broad Commodities	2026-06-03	7	-8.5%	0.99	-0.97
USO	cluster_20	Broad Commodities	2026-06-03	7	-11.0%	1.00	-0.99
PDBC	cluster_20	Broad Commodities	2026-06-03	7	-5.5%	1.00	-1.00
DBC	cluster_20	Broad Commodities	2026-06-03	7	-5.7%	1.00	-1.00
DJP	cluster_20	Broad Commodities	2026-06-03	7	-6.4%	1.00	-1.00
SIVR	cluster_17	Gold	2026-05-27	12	-9.2%	1.00	-1.00
SLV	cluster_17	Gold	2026-05-27	12	-9.2%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
KBWB	cluster_14	Financials	1.00	1.00
RSPH	cluster_11	Healthcare	1.00	1.00
RPV	cluster_8	U.S. Dividend Equity	1.00	1.00
PEJ	cluster_2	U.S. Mid Cap Equity	1.00	1.00
ONEV	cluster_3	U.S. Large Cap Value	1.00	1.00
PSCC	cluster_18	Consumer Staples	1.00	1.00
NURE	cluster_15	Real Estate	1.00	1.00
HEZU	cluster_4	Developed ex-US Equity	1.00	0.99
SHM	cluster_16	Municipal Bond	0.98	0.98
SPHQ	cluster_1	U.S. Large Cap Equity	1.00	0.90
IAGG	cluster_5	Aggregate Fixed Income	0.97	0.85
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.90	0.83
IPO	cluster_6	U.S. Large Cap Growth	1.00	0.64
PCY	cluster_10	High Yield Bond	0.64	0.54
EWT	cluster_7	Emerging Markets Equity	1.00	0.50
ENZL	cluster_9	Asia-Pacific Equity	0.72	0.21
DBJP	cluster_22	Japan Equity	0.97	0.11
FLTR	cluster_23	Bank Loan / Floating Rate	1.00	0.04
SLX	cluster_12	Materials	0.63	-0.28
VTIP	cluster_21	Inflation-Protected Fixed...	0.31	-0.51

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
EWI	cluster_13	China Equity	1.00	-1.00
DBA	cluster_20	Broad Commodities	1.00	-1.00
IAU	cluster_17	Gold	1.00	-1.00
VNM	cluster_19	India Equity	1.00	-1.00
FXC	cluster_9	Asia-Pacific Equity	1.00	-1.00
MOO	cluster_12	Materials	1.00	-1.00

FXF	cluster_25	International Fixed Income	1.00	-1.00
FCOM	cluster_1	U.S. Large Cap Equity	1.00	-1.00
SRLN	cluster_23	Bank Loan / Floating Rate	1.00	-1.00
IPKW	cluster_4	Developed ex-US Equity	1.00	-1.00
TDTT	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
PNQI	cluster_6	U.S. Large Cap Growth	1.00	-1.00
SDIV	cluster_7	Emerging Markets Equity	1.00	-1.00
PSK	cluster_10	High Yield Bond	1.00	-1.00
PSP	cluster_14	Financials	1.00	-1.00
SPTS	cluster_5	Aggregate Fixed Income	0.97	-0.96
NEAR	cluster_24	Cash / Ultra-Short Fixed ...	0.95	-0.95
IPAC	cluster_22	Japan Equity	1.00	-0.87
PTH	cluster_11	Healthcare	0.78	-0.77
FUTY	cluster_18	Consumer Staples	0.71	-0.71

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_19	11	FLAT	0.304	0.094	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.488	0.077	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.716	0.041	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.731	0.094	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.615	0.218	0.00036	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_20	10	FLAT	0.598	0.172	0.00040	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.134	0.00031	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.661	0.187	0.00046	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.162	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.706	0.192	0.00044	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.025	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.083	0.00021	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_19	11	FLAT	0.499	0.219	0.00055	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.755	0.181	0.00044	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	FLAT	0.845	0.127	0.00035	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.876	0.161	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.819	0.203	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.837	0.238	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_19	11	FLAT	0.445	0.275	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.878	0.249	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_19	11	FLAT	0.445	0.330	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.264	0.00053	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.803	0.270	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	FLAT	0.750	0.360	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_19	11	FLAT	0.505	0.358	0.00103	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.866	0.327	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_19	11	FLAT	0.563	0.410	0.00107	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.856	0.258	0.00072	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.228	0.00056	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.904	0.242	0.00064	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20